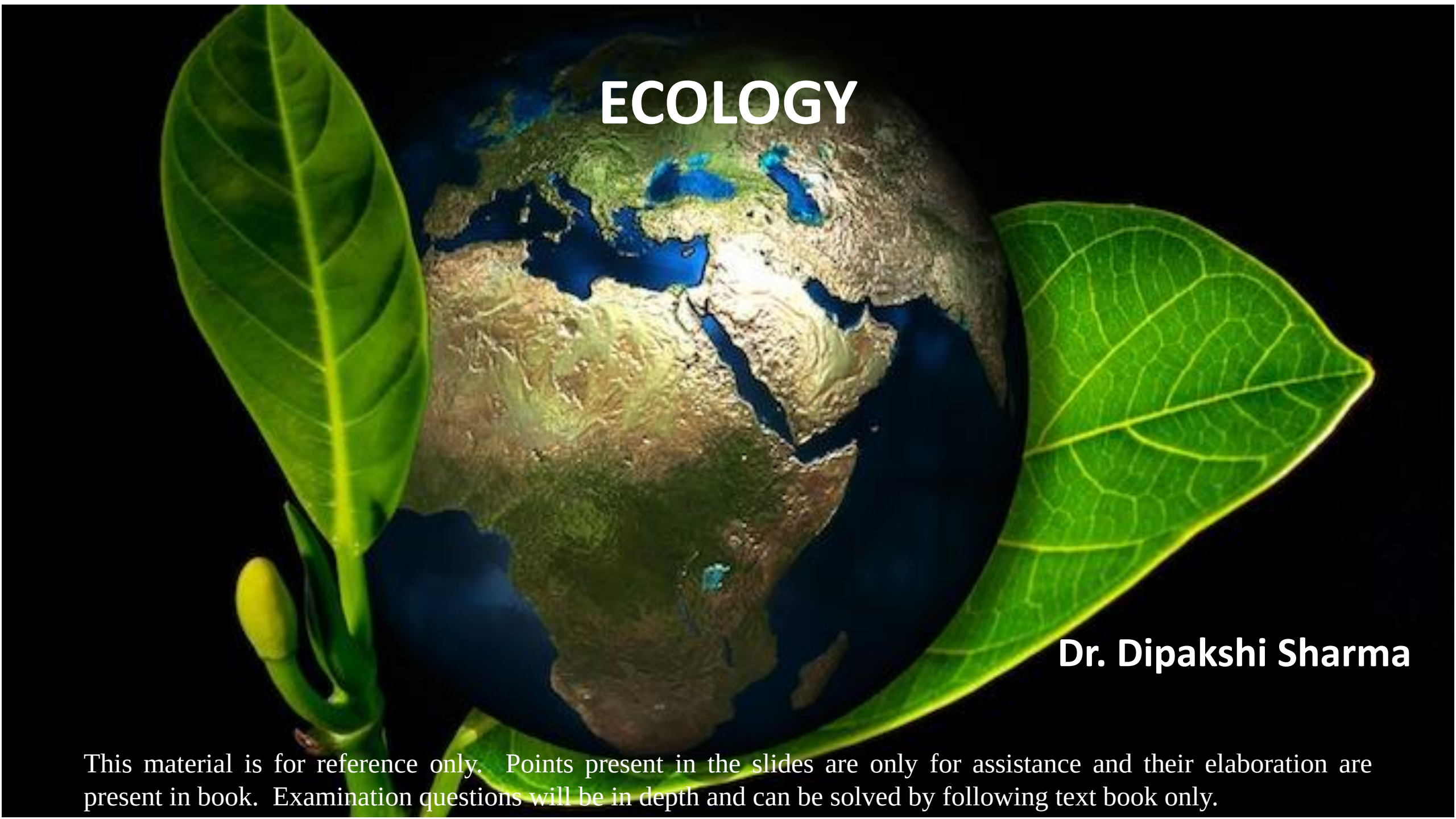

ECOLOGY

Dr. Dipakshi Sharma

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Introduction-Ecology

The term **Oekologie** (Ecology) was coined by the German biologist **Ernst Haeckel** in the year 1866. Greek word "Oikos" meaning "Home" and "Logos" meaning "Study"

Ecology:- The study of relationship of various organisms in their natural habitat interacting with their surroundings.

- **Autecology-**--Study of single species/organisms.
- **Synecology-**--Study of multiple species/ group of organism.

Ecosystem

The term **Ecosystem** was first used in 1935 in a publication by British ecologist **Arthur Tansley**.

- It is defined as a basic structural and functional unit of Ecology.
- Each ecosystem has a definite structure and definite role to play in functioning of that place specific environment. eg. Grassland ecosystem, forest ecosystem, aquatic ecosystem etc.

- A self-regulating group of biotic communities of species interacting with one another and with their non-living environment exchanging energy and matter.

Classification of Ecosystem

1. Natural Ecosystem
   - Aquatic
     - Fresh water
     - Running water
     - Standing water
   - Marine
   - Terrestrial
     - Grassland
     - Forest
     - Desert
2. Artificial/Engineered Ecosystem

Structural units/Components of an Ecosystem

Biotic Components

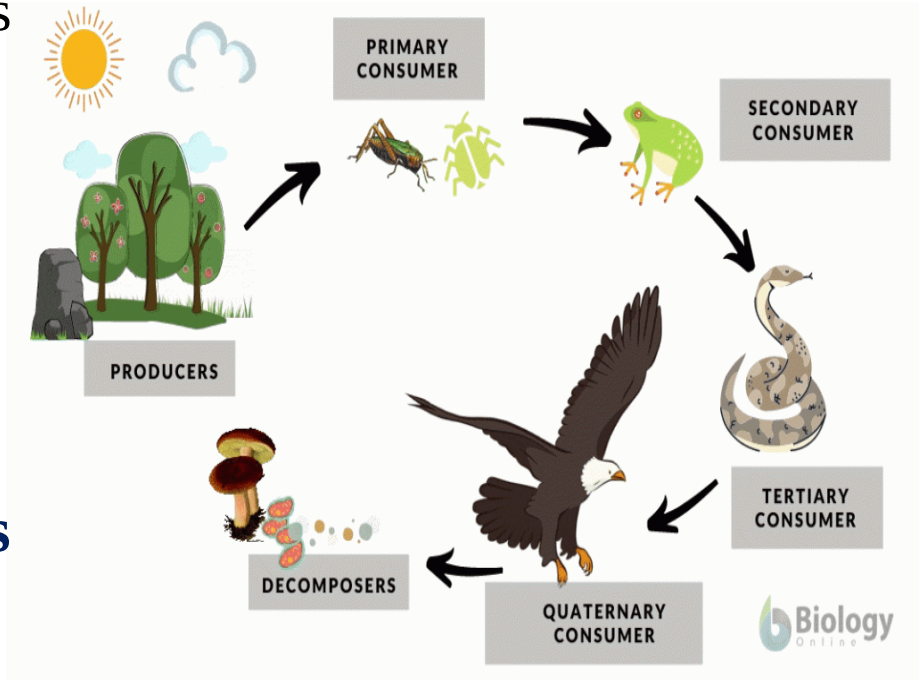
1. Producers

- Photo-autotrophs
- Chemo-autotrophs- *Nitrosomonas*, Iron bacteria, Methanogens

2. Consumers

- Herbivores
- Carnivores
- Omnivores
- Detrivores

3. Decomposers



Abiotic Components

1. Physical

- **Climatic-** Sunlight, Temperature, Humidity, Rainfall, Wind etc.
- **Edaphic-** Soil type, Soil moisture, Soil reaction.
- **Geographic-** Latitude, Longitude, Altitude

2. Chemical

- Macro/micro nutrients
- Trace elements
- Pollutants
- Organic substances

Functional units/Components of an Ecosystem

- **Food Chain**
- **Food Web**
- **Trophic Levels**
- **Energy Flow**

Biogeochemical Cycles

1. Gaseous Cycles

- C, N, O and Hydrological Cycle

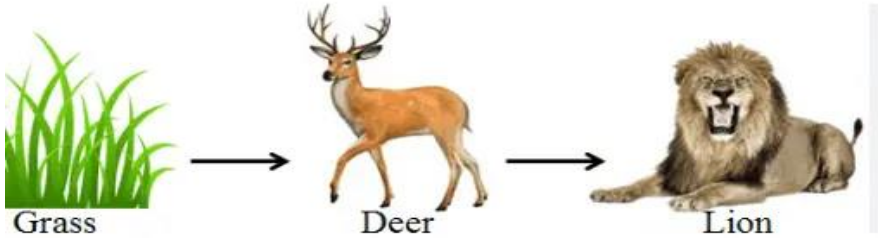
2. Sedimentary Cycles

- Phosphorus Cycle
- Sulphur Cycle

Food Chain

The linkage between the living organisms (plants/animals) for the transfer of food energy is known as food chain.

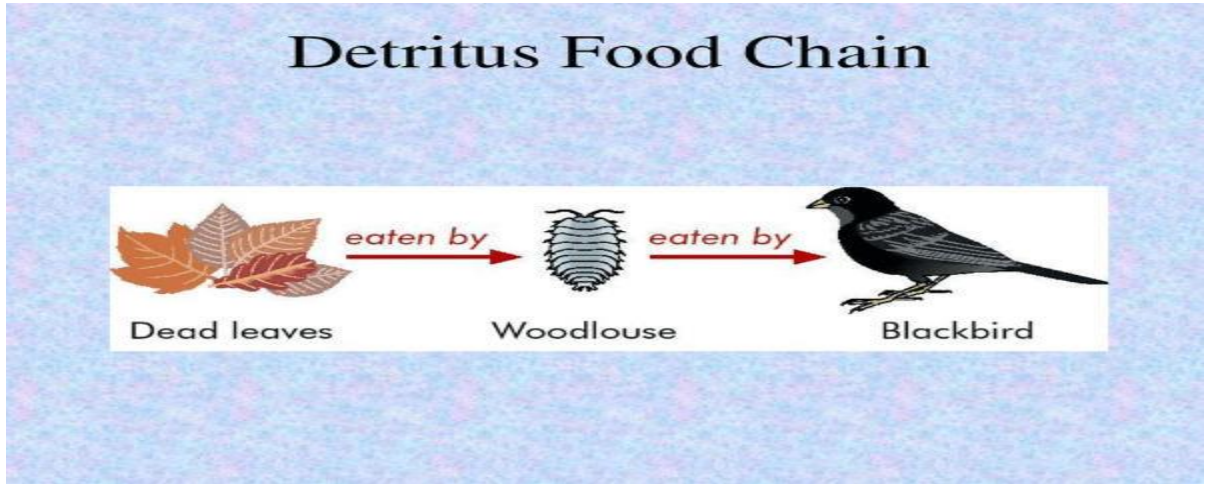
- Food chains are not simple and straight.

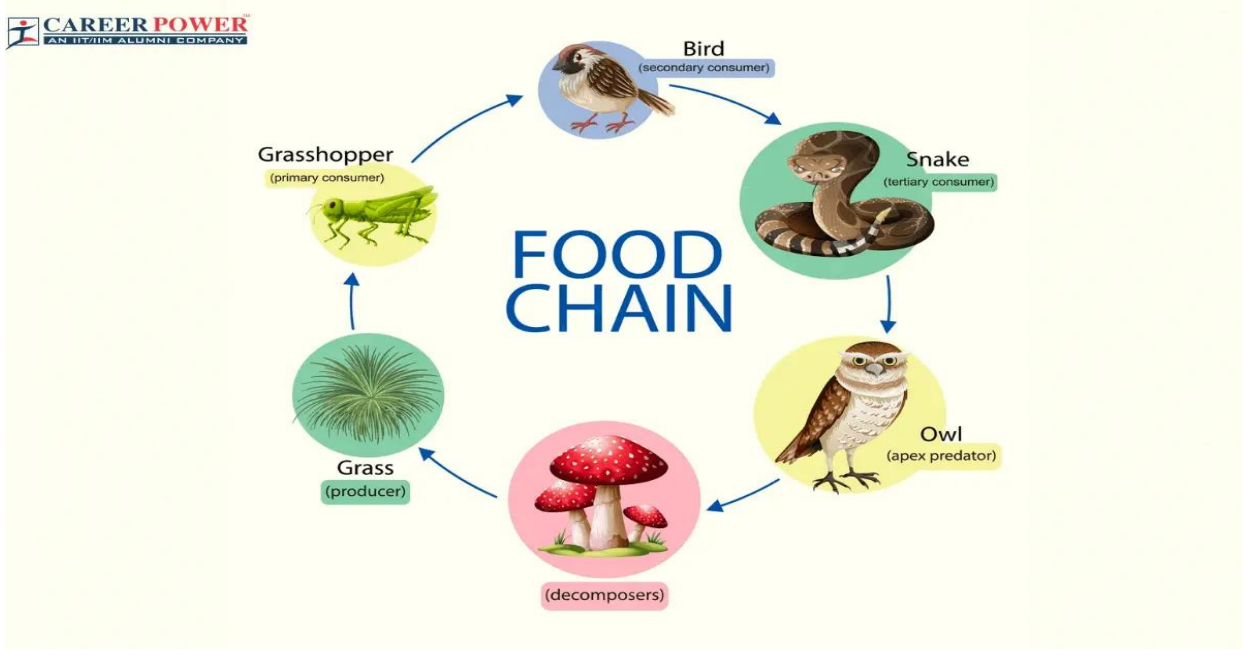


Types of Food Chains

1. Grazing food chain- Energy flow begins with plants fixing solar light.

2. Detritus food chain- Energy flow begins with dead organic matter broken down by decomposers.





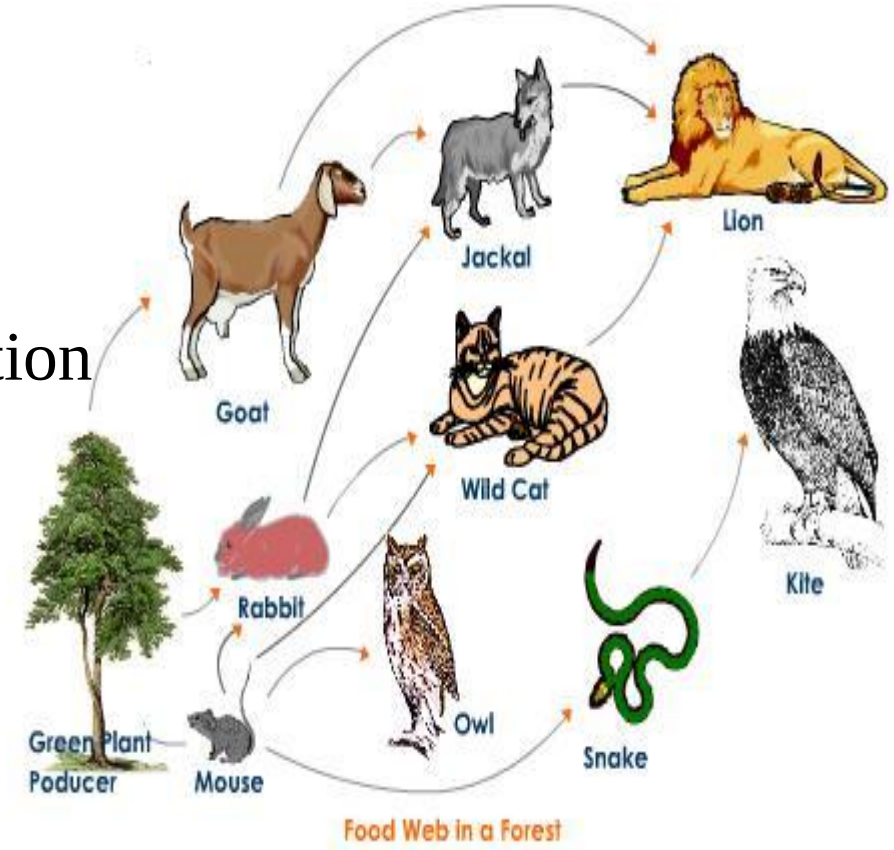
Food Web

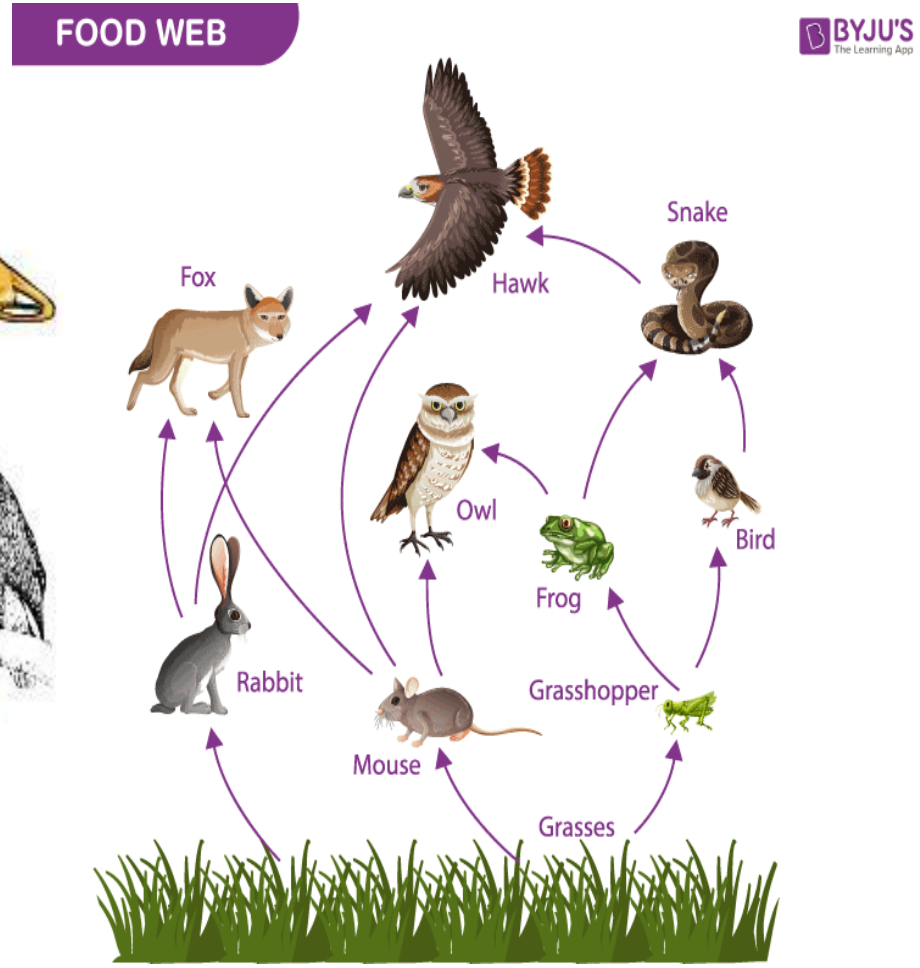
The inter linkage between different food chains is known as food web.

Two food chains involving same number of steps are said to be belonging to same trophic level.

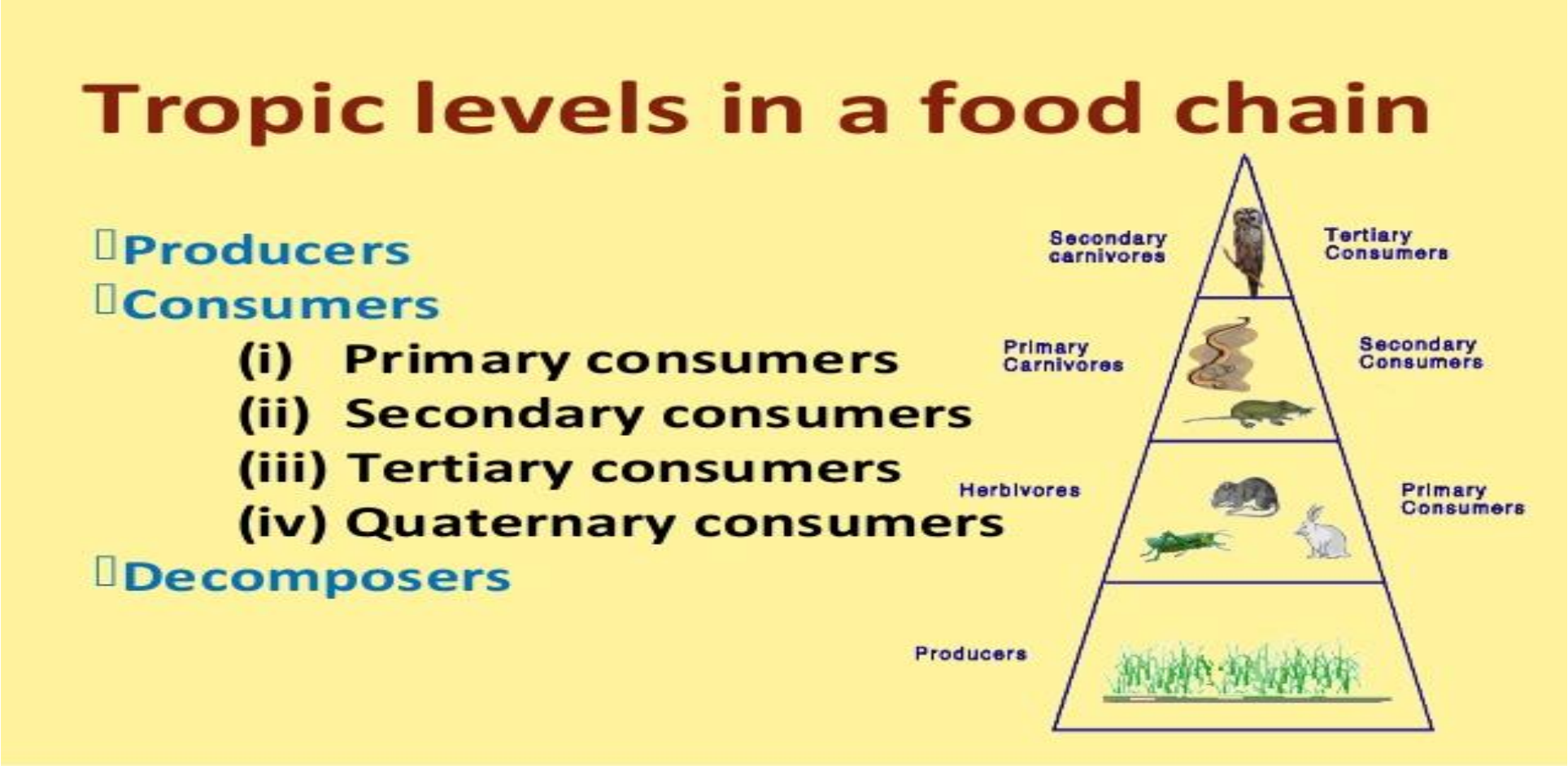
Significance of Food chain/Food web

- Energy flow
- Nutrient cycles
- Ecological balance
- Population size regulation





Trophic Level- The various steps in a food chain at which the transfer of food or energy takes place are called trophic levels.



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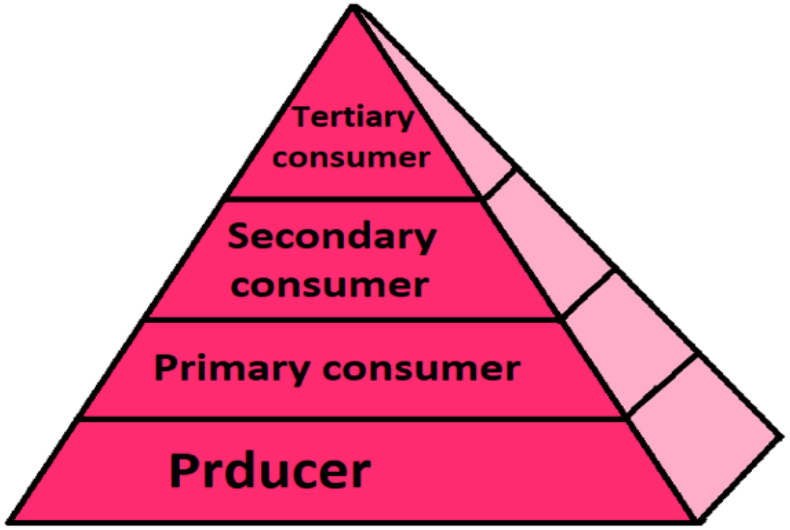
Ecological Pyramids

An ecological pyramid depicts how energy and biomass decrease from lower to higher trophic levels.

- The producer or first trophic level makes up the **base** of the pyramid. The successive three levels make the **apex**.

Types of Ecological Pyramid

1. Pyramid of Numbers
2. Pyramid of Biomass
3. Pyramid of Energy



Pyramid of Numbers

This pyramid depicts the relationship between herbivores, producers and carnivores at successive trophic levels in terms of their numbers.

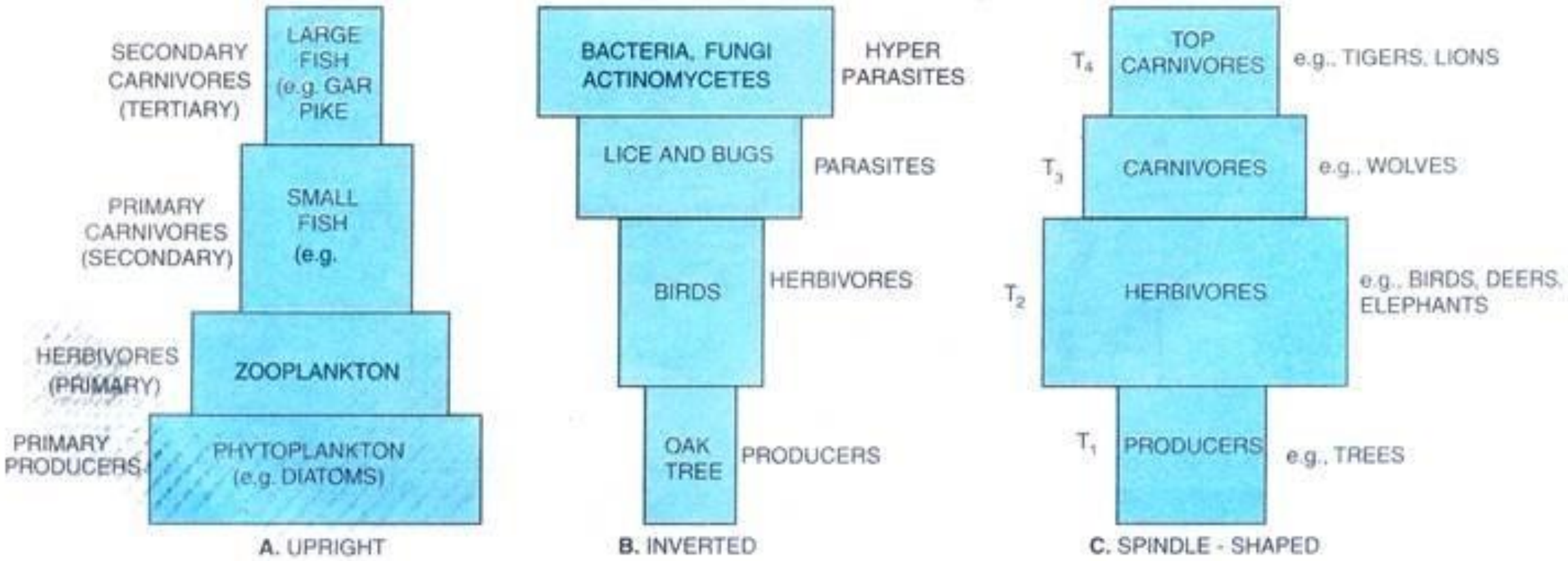


Fig. 14.15. Pyramids of numbers : **A.** In pond ecosystem; **B.** In parasitic food chain; **C.** Tree ecosystem.

Pyramid of Biomass

This Pyramid depicts the quantitative relationships of the standing crops instead of the geometric factor.

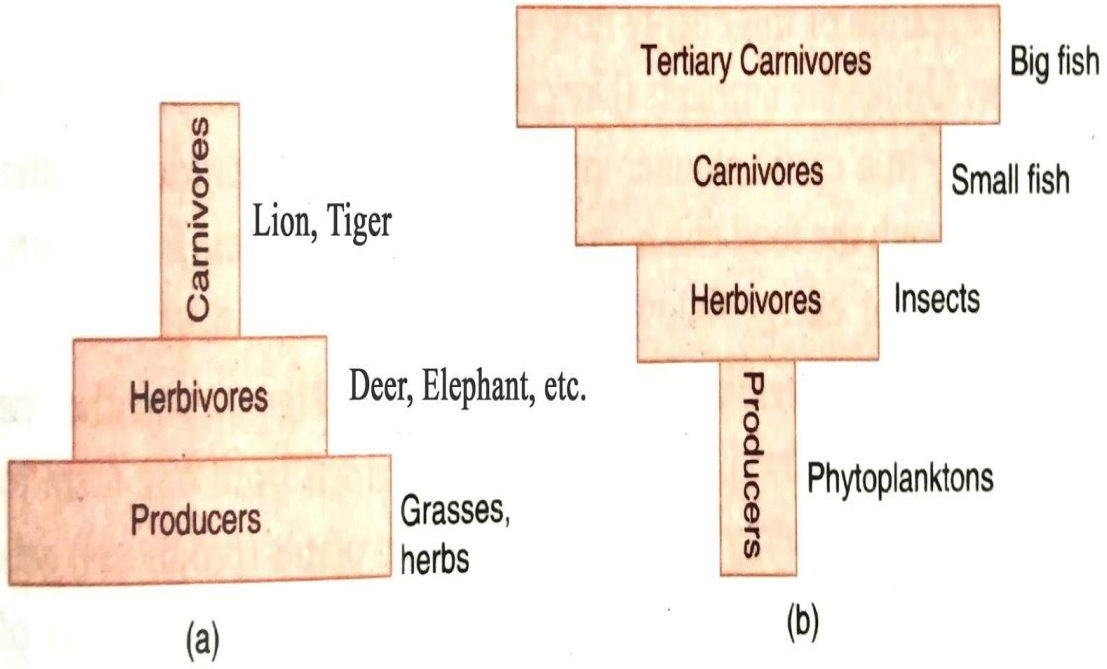
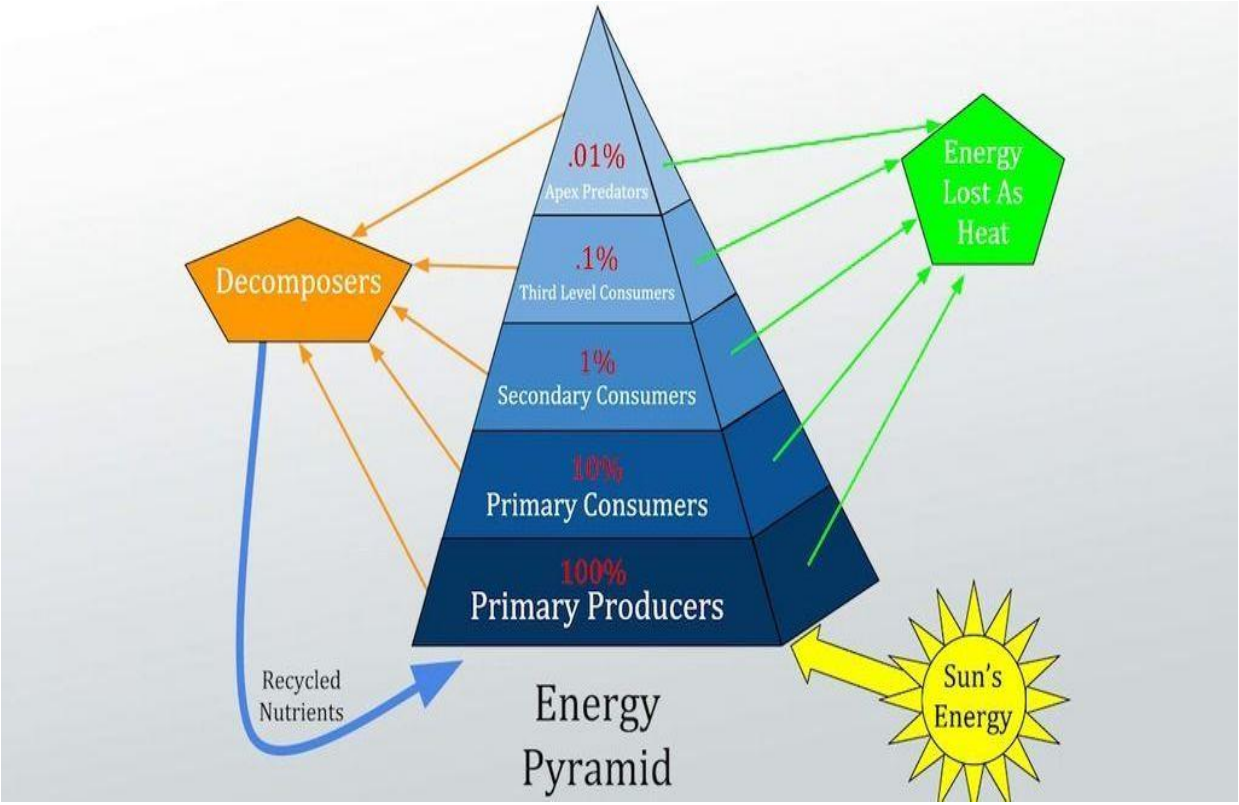


Fig. Pyramid of biomass (*a*) Forest (*b*) Pond.

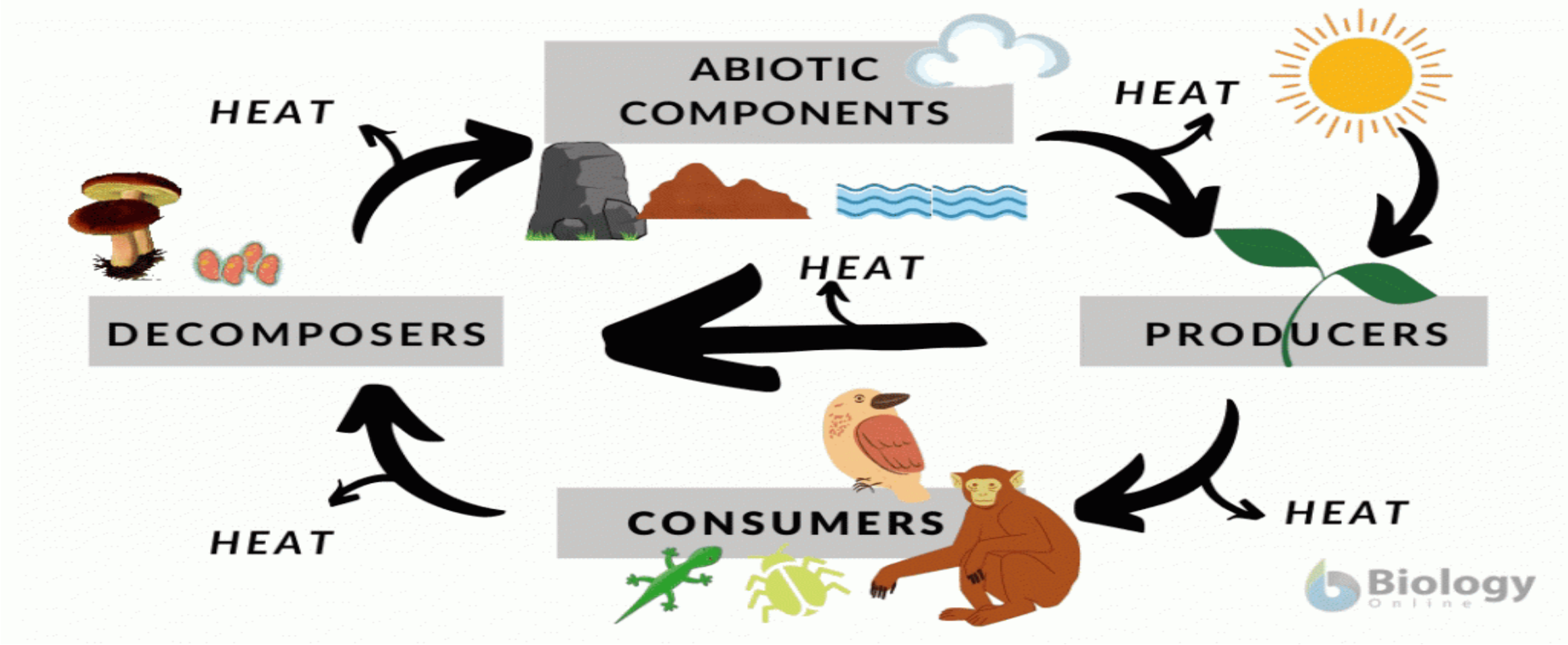
Pyramid of Energy

This Pyramid depicts the weight and number of organisms at any level are based on the rate at which food is being produced. It is upright in shape.



Energy Flow in Environment

Energy flow in the environment is uni-directional



Energy flow

- Universal energy flow model

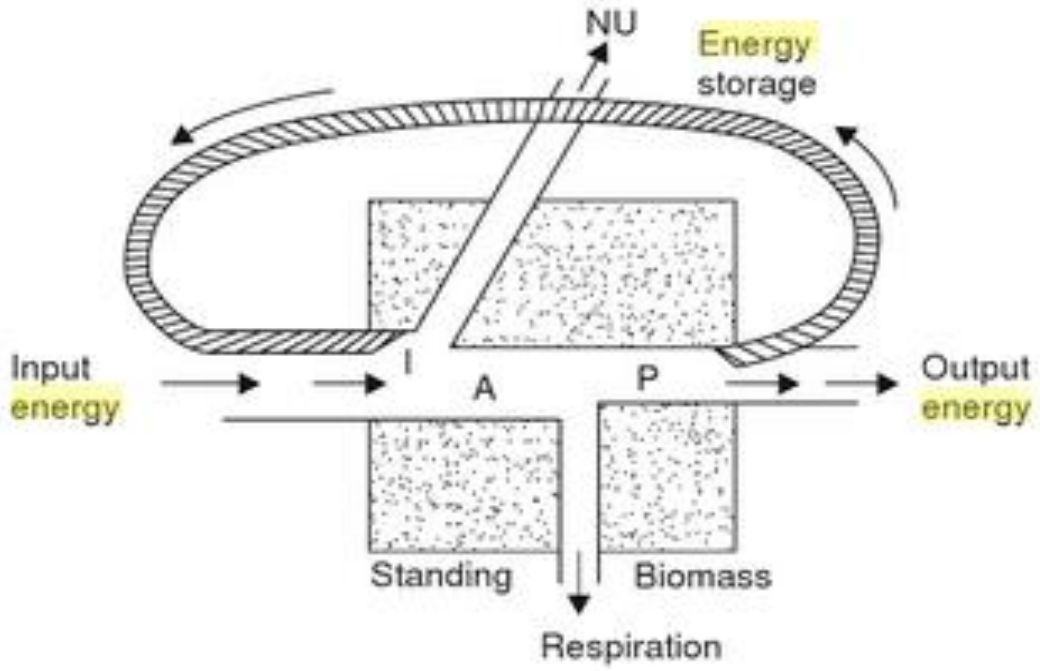


Fig. 3.8. Universal energy flow model applicable to all living components (I = Energy input; A : assimilated energy ; P = Production ; NU = Energy not used.

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Energy flow

- Single channel energy flow model

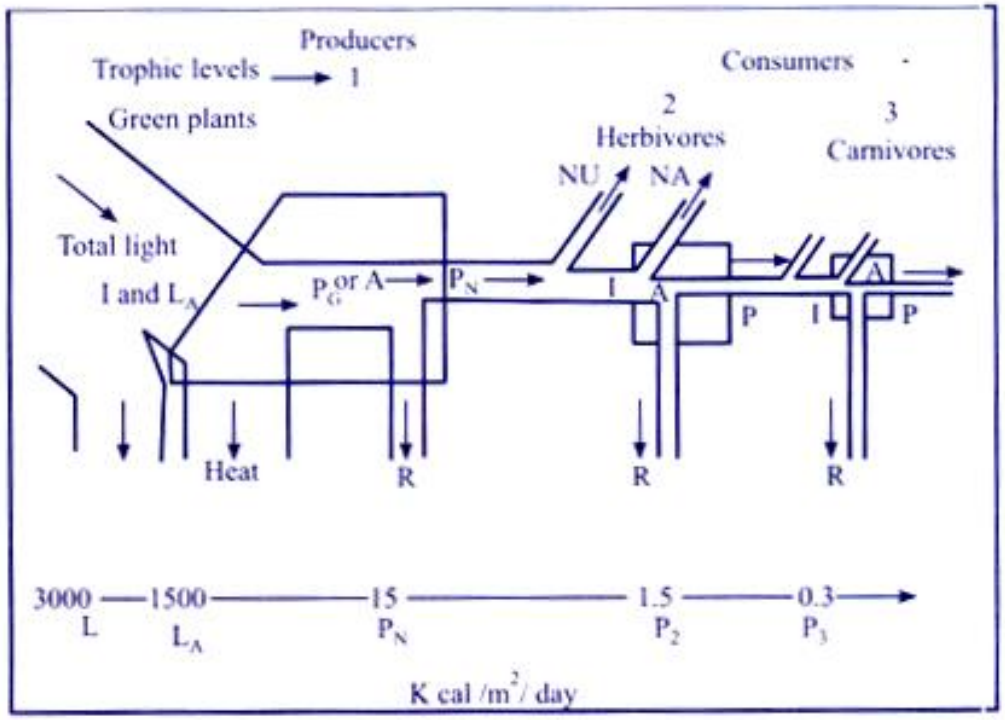

Fig. 1.4 A simplified energy flow diagram depicting three trophic levels

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Energy flow

- Double channel or Y-shaped energy flow model

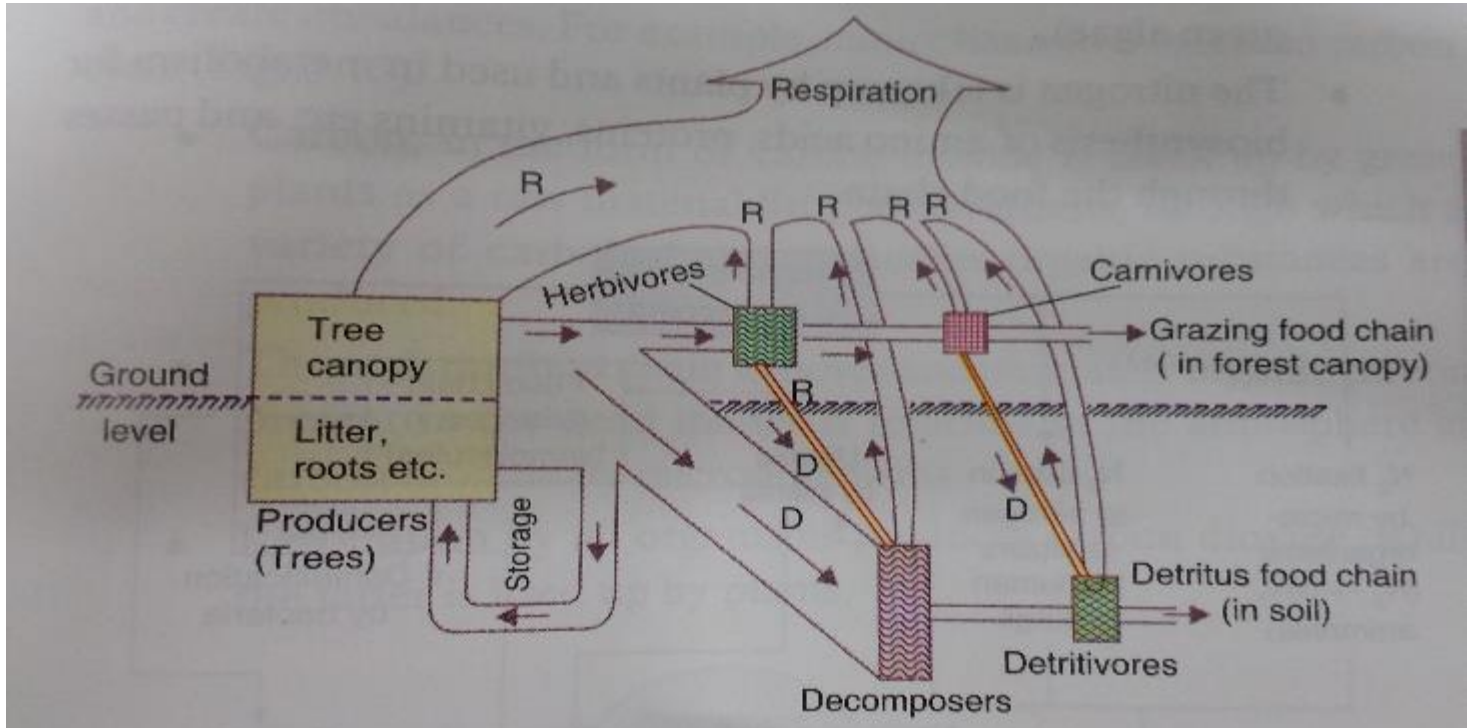

Fig. 3.13 Y-shaped or 2-channel energy flow model showing energy flow through the grazing food chain and the detritus food chain (R = Respiration, D = Detritus or dead matter).

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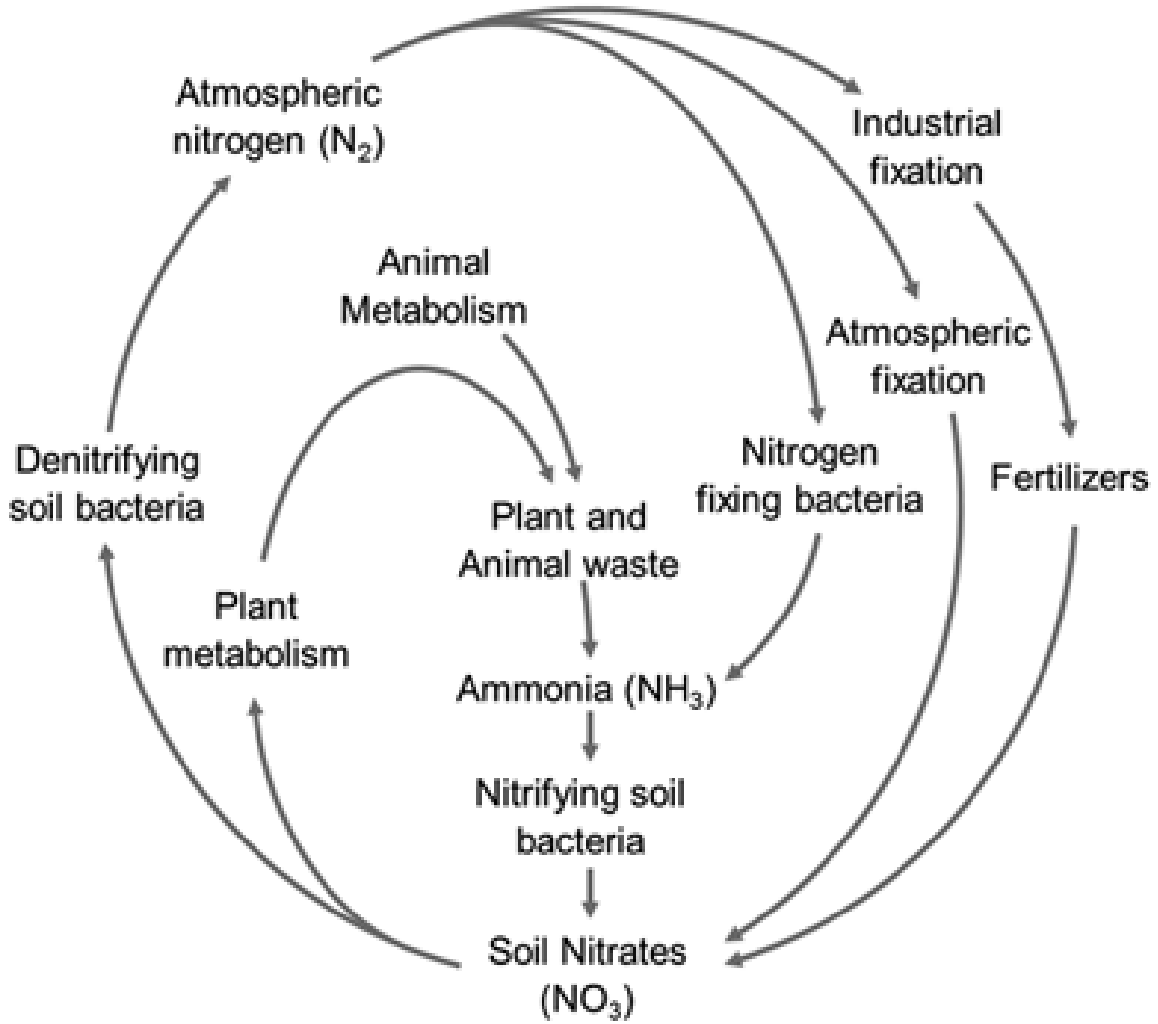
Nutrient Cycles

Nitrogen



Carbon

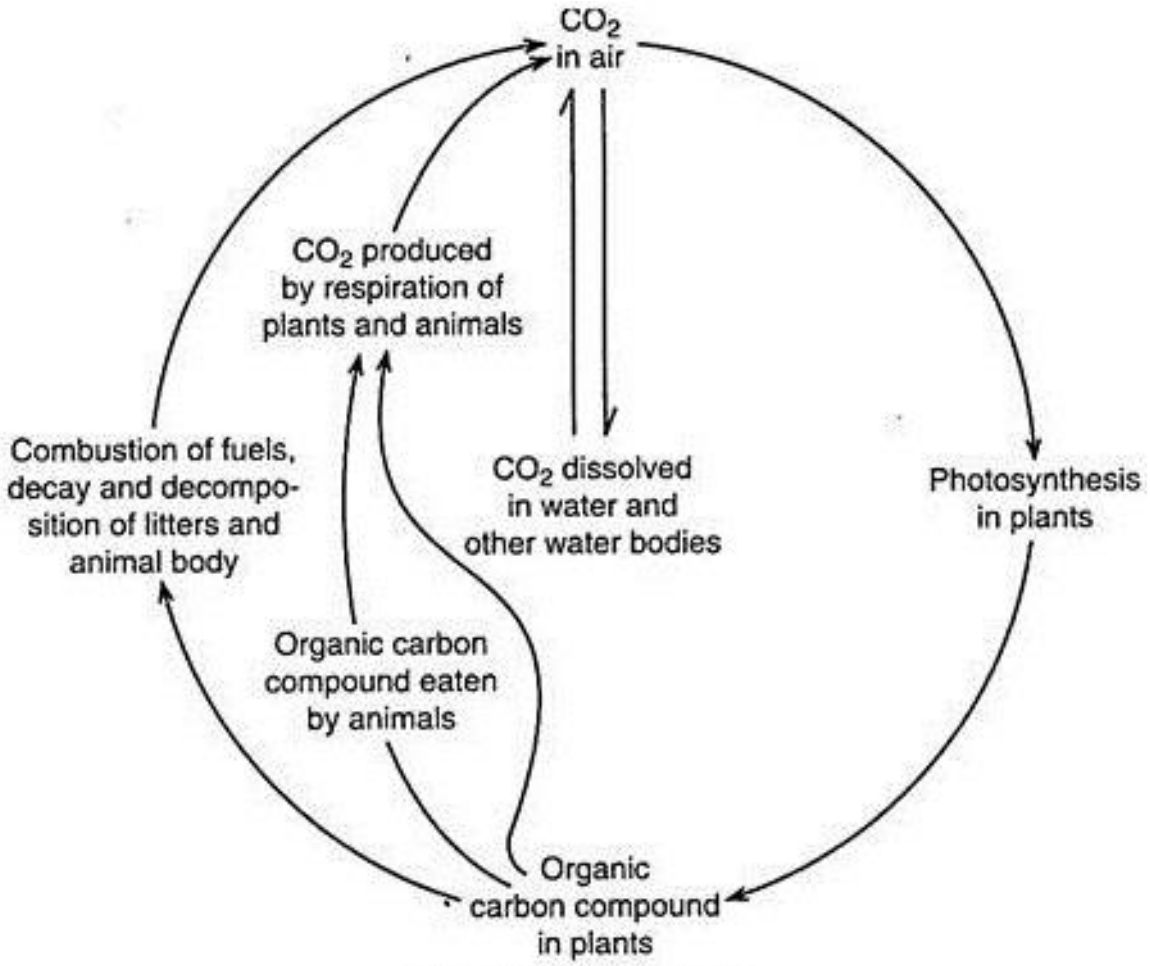
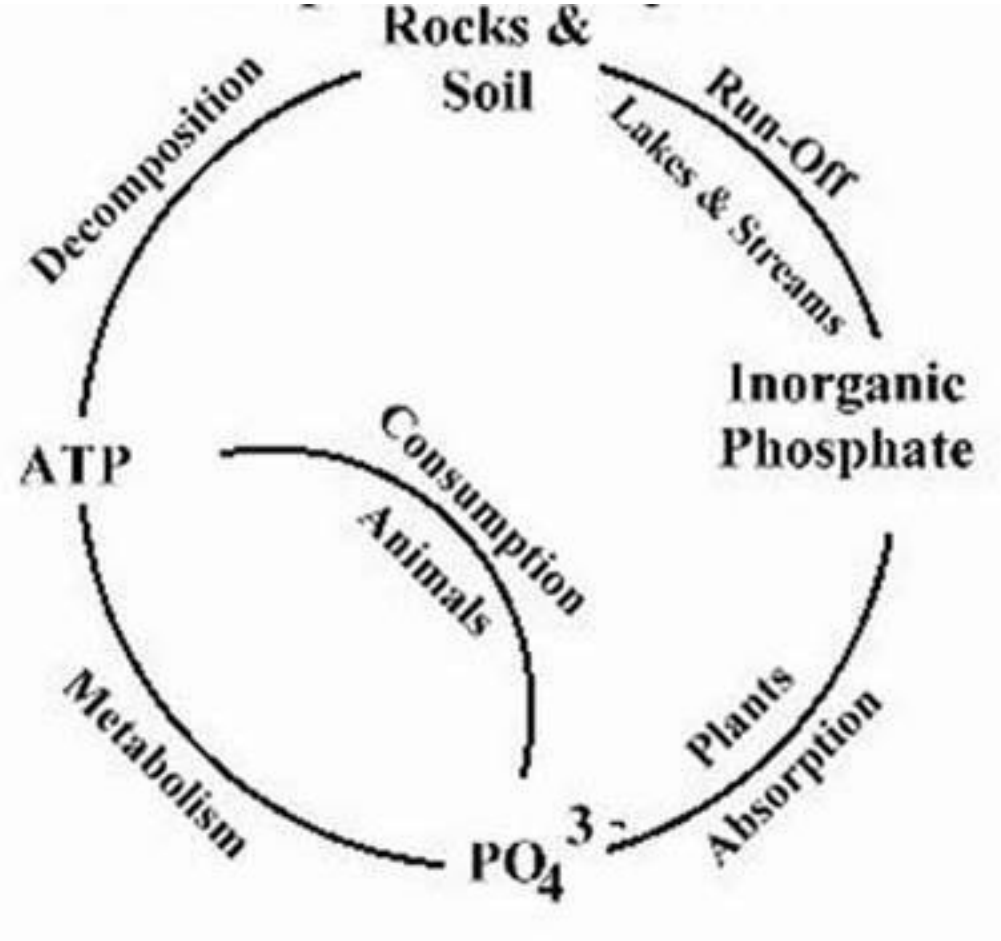


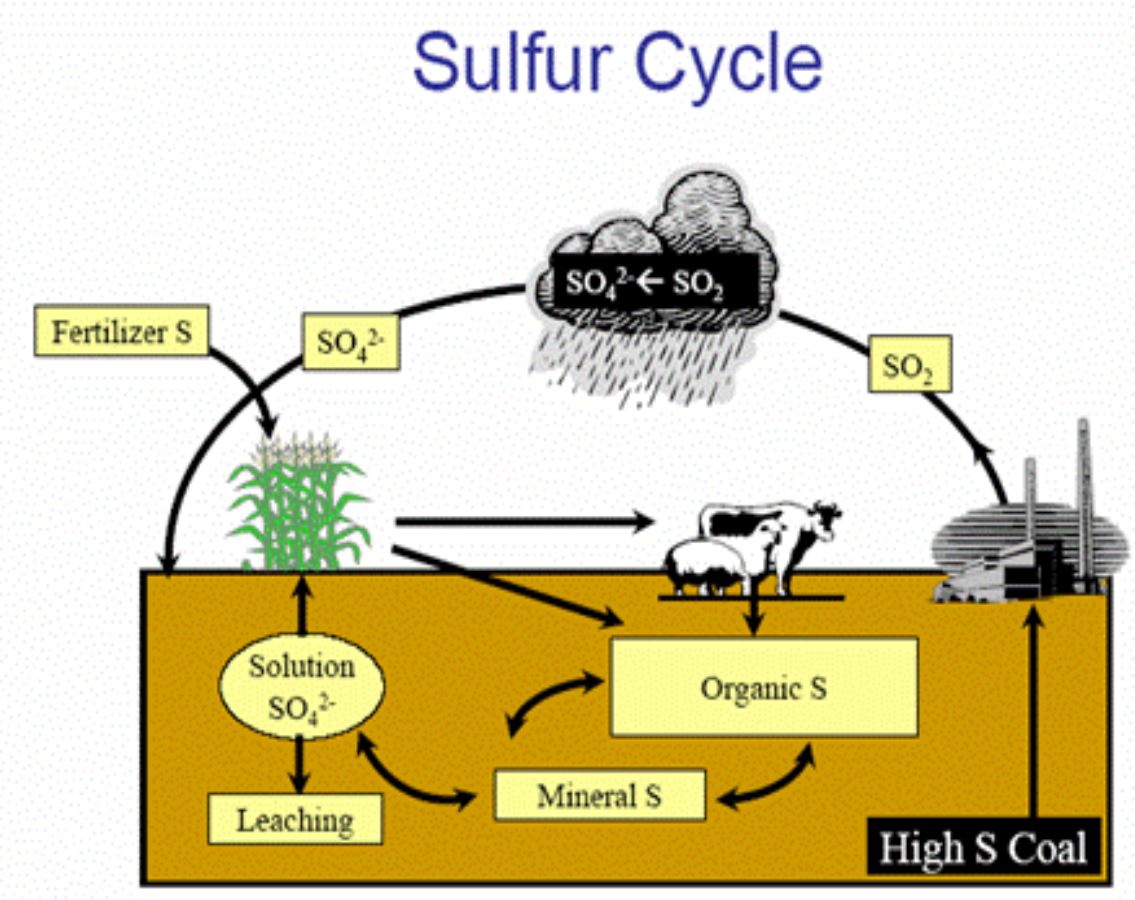
Fig. 2.33 : Carbon cycle

Nutrient Cycles

Phosphorus



Sulphur



Production of biomass

- Primary production
  - Biomass production using photosynthesis
- Secondary production
  - Biomass production by consuming producers

Homeostasis

It means resistance to changes in the external environment

or

The tendency to maintain a stable, relatively constant internal environment is called **homeostasis**.

The term given by Claude Bernard, a French Physiologist in the year 1865 and was first used in 1926 by Walter Bradford Cannon.

The regulation of homeostasis depends on three mechanisms:

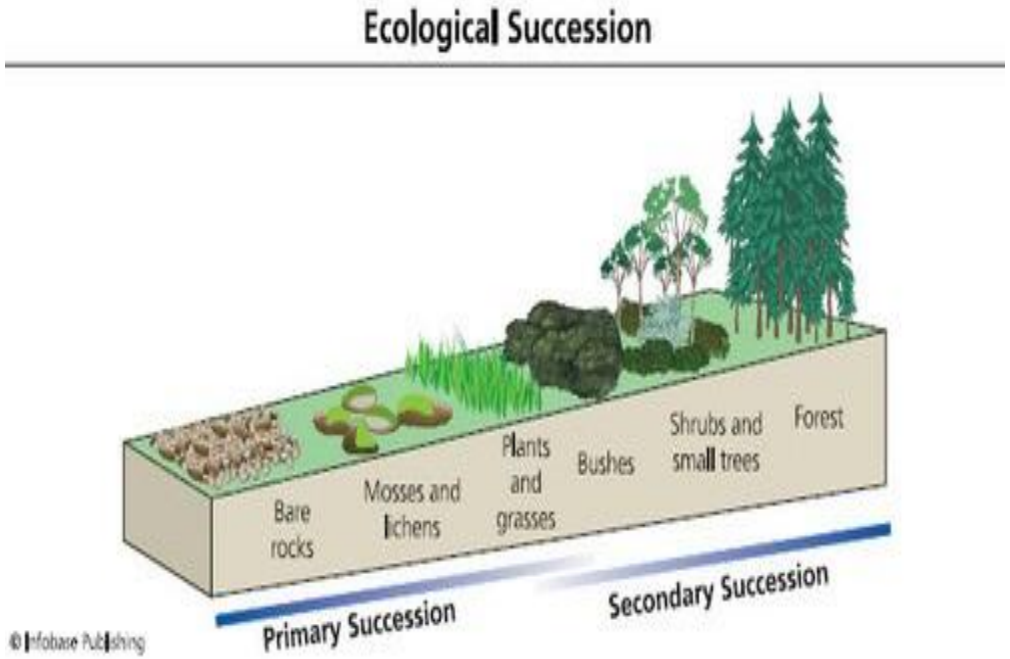
- Effector
- Receptor
- Control Center

1. Positive feedback: Stimulates change and it is responsible for the sudden appearance of rapid changes within ecosystems.
2. Negative feedback: Maintenance of homeostasis usually involves negative feedback loops.

Ecological Succession

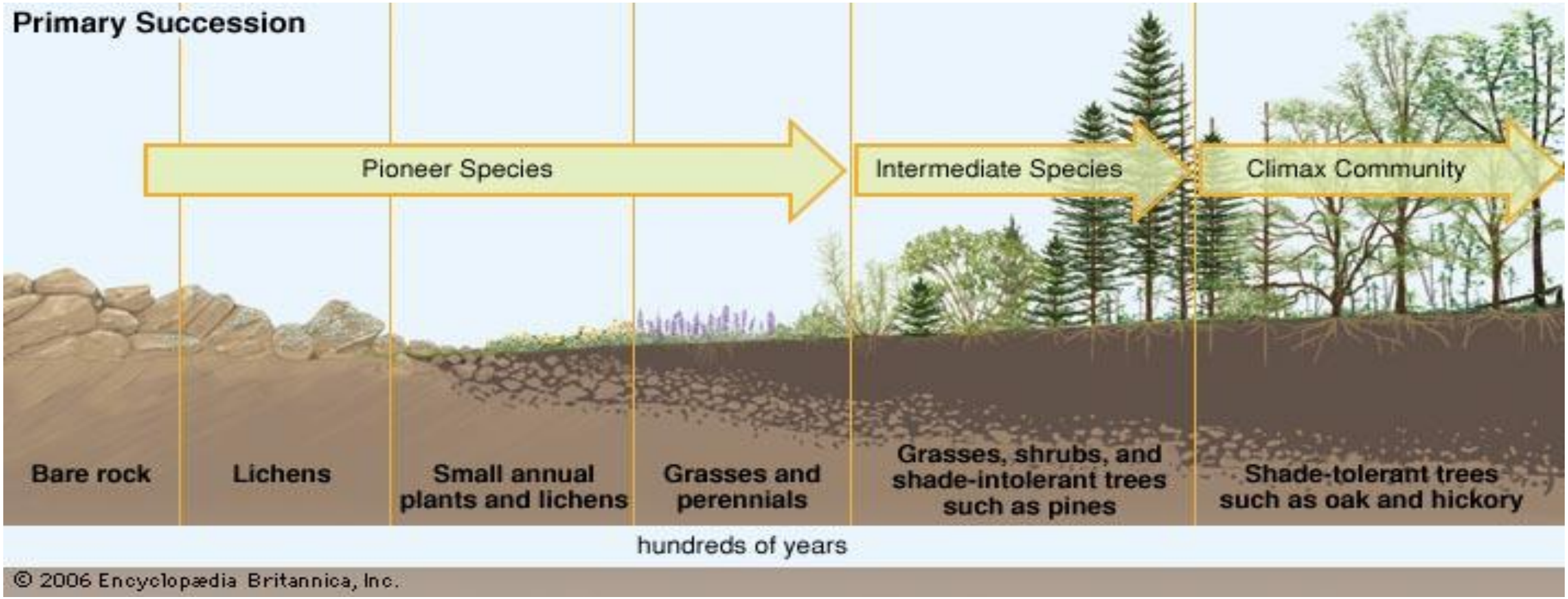
Ecological succession is the gradual process by which **ecosystems change and develop over time.**

- It is therefore a series of predictable temporary communities or stages leading up to a climax community.
- Each stage/temporary community is called a successional stage or seral stage.
- Each step prepares the land for the next successional stage.
- All habitats are in the state of constant ecological succession.

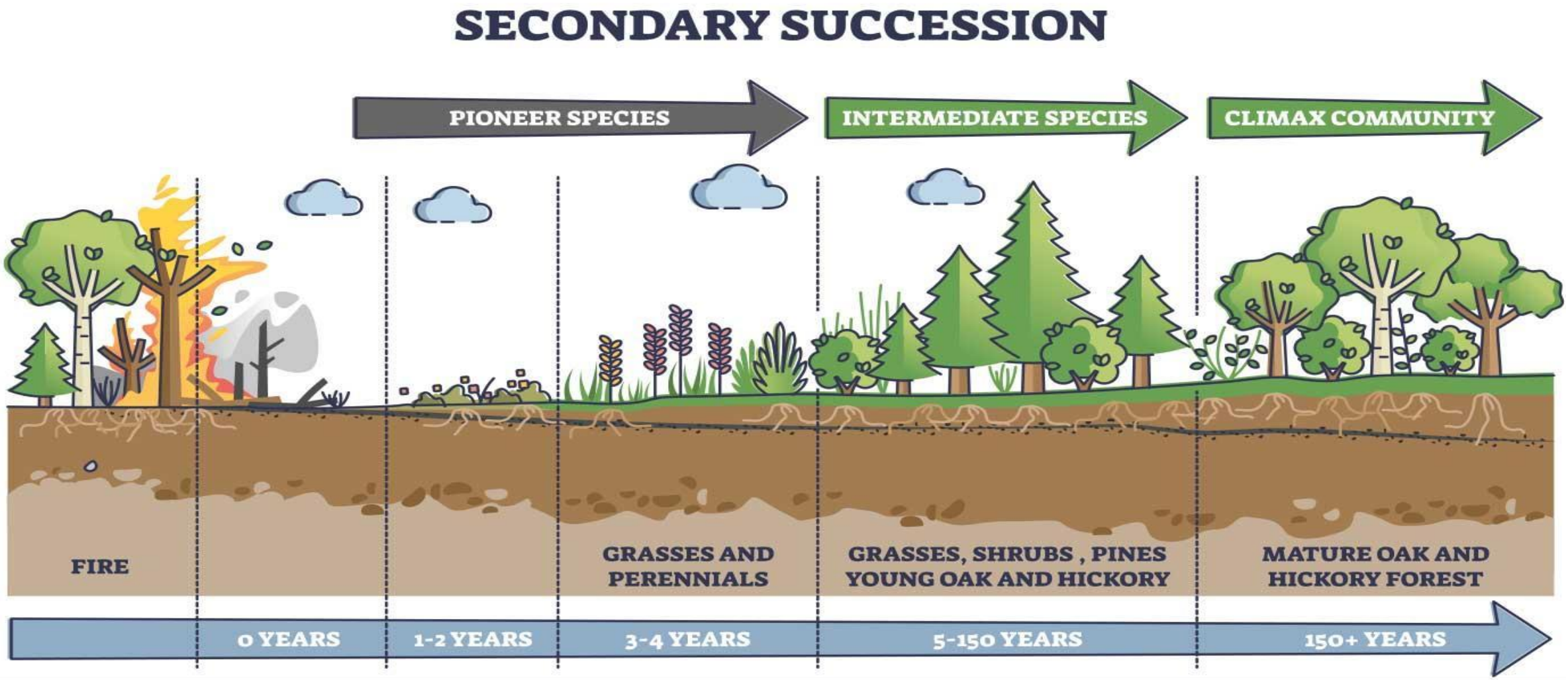


Types of Succession

Primary succession: refers to a series of community changes which occur on an entirely new habitat which has never been colonized before. For example, a newly quarried rock face or sand dunes. (pioneer and climax community).



Secondary succession: refers to a series of community changes which take place on a previously colonized, but disturbed or damaged habitat. For example, land obtained after felling trees in a woodland, land clearance, or fire.



Classification of Succession at different type of area

Hydrarch / Hydrosere: Pond, swamp, bog

Mesarch: Area with adequate moisture

Xerarch / Xerosere: Dry lands

Lithosere: On bare rock

Psammosere: On sand

Halosere: On saline soil

Process of Ecological Succession

1. **Nudation**: The formation of a bare area or nudation by several reasons, such as volcanic eruption, flooding, landslide, erosion etc.
2. **Invasion**: Arrival of the reproductive bodies or propagules of various organisms and their settlement in the new or bare area.
   - Migration (Dispersal)
   - Ecesis (Establishment)
   - Aggregation
3. **Competition and Co-action**: Increase in number of species with the limited space.
4. **Reaction**: Modification of the environment through the influence of the living organism present on it. Reaction causes change in land, soil, water, temperature of the area.
5. **Stabilization**: Final or terminal communities become more or less stabilized for a longer period of time. It can maintain its equilibrium with the climate of the area. Final community is called the climate community which is determined by the climate of the region. Examples of climax communities are – grassland, forests, coral reef.

Forest ecosystem

Abiotic Components

- Inorganic and organic substances found in soil
- Temperature, humidity, rainfall, light
- Biogenic gases (CO_2, O_2)

Biotic components

Producers: Large trees, herbs, lianas (climbers), Orchids

Primary consumer: Deer, Elephant, moles etc.

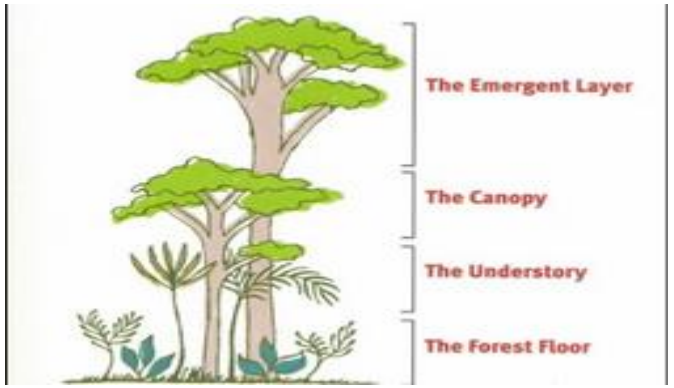
Secondary consumer: Snake, Lizards etc.

Tertiary consumers: Tiger, Lion etc.

Decomposers

Tropical rain forest

- Notable features:
  - Found in tropical region (near the equator).
  - High rainfall, humidity and temperature
  - Large leaves
  - Rich in biodiversity
  - Fauna of these rainforests includes the jaguar, tapir, okapi, boa constrictor, African grey parrot, keel-billed toucan, crowned eagle, three-toed sloth, spider monkey, large flying fox and more.
- Layers
  - Emergent layer
  - Canopy
  - Understory
  - Forest floor
- Example: Amazon Rainforest, Congo Rainforest, Southeast Asian Rainforest etc.



Tropical Deciduous Forest

Notable features:

- Tropical deciduous forests form a natural cover almost all over India.
- They are of two types-moist and dry. Moist forests are found on the eastern slopes of Western Ghats, North eastern parts of the Peninsular Chhota Nagpur plateau and along the Shiwaliks.
- They shed there leaves for a particular period of time.
- They are economically very important because of timber like sal and teak.
- The animals found here are tiger, wolves, rabbit etc.

Sub-Tropical Forest

Notable features:

- Subtropical forests are within or bordering the tropical zone.
- temperatures may vary only slightly over a year
- Subtropical rainforests occur in Central America, the West Indies, India, Madagascar, mainland Southeast Asia, and the Philippines.
- Small deciduous trees and shrubs are found.

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Temperate Rain Forest

Notable features:

- The world's largest temperate rain forests are found on the Pacific coast of North America.
- Temperate rain forests are also found in coastal Chile, Norway, the United Kingdom, Japan, Australia and New Zealand.
- The mild weather conditions
- Adequate rainfall
- Coniferous trees dominate the forest
- tall evergreen trees are also found
- Animals: black bears, lynx, wolves etc.

Temperate Deciduous Forest

Notable features:

- Located in the mid-latitude areas (between the polar regions and the tropics).
- The temperature varies widely from season to season with cold winters and hot, wet summers.
- During the fall, trees change color and then lose their leaves.
- Most of the trees are broadleaf trees such as oak, maple, beech, hickory and chestnut.
- Animals: toad, chipmunk, gray squirrel, Yellow-breasted chat etc.

Evergreen Coniferous Forest

Notable features:

- They are found just in south of arctic tundra
- Winters are long, cold and dry
- Sunlight is available for few hours only
- Soil has less nutrient and acidic
- Major trees are Pine, Fir, Cedar etc.
- Animals: moose, deer, reindeer, squirrels, wolves, bears, foxes, owls, woodpeckers hawks etc.

Grassland Ecosystem

- Abiotic components:
  - Inorganic elements (C, H, O, N, P, S)
  - Temperature, humidity, rainfall, light
- Biotic components:
  - Producers: Some scattered trees, Grass
  - Primary consumers: Grazing animals,

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Tropical Grassland

Notable features:

- Located near the equator, between the Tropic of Cancer and the Tropic of Capricorn.
- Although these areas are overall very dry, they do have a season of heavy rain.
- Dominated by grasses
- May have some drought-resistant, fire-resistant trees
- Animals: Giraffes, zebras, buffaloes, kangaroos, mice, moles, gophers, ground squirrels, snakes, worms, termites, beetles, lions, leopards, hyenas, and elephants.

Examples: Savannas

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Temperate Grassland

Notable features:

- Trees and shrubs are completely absent or rare.
- cold winters (-40ºC) and hot summers (38ºC)
- Gentle slope
- Animals: Rodents, bisons, wolves, hawks, owls etc.

- Examples:
  - Pampas- South America
  - Veldts- South Africa
  - Steppes- Asia and Europe
  - Downs- Australia

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Polar Grassland

Notable features:

- Severe cold and strong wind
- Arctic wolf, arctic fox, reindeer, migratory birds and insects are found.

Examples: Arctic Tundra

- **Permafrost:** Permafrost is soil, rock or sediment that is frozen for more than two consecutive years. In areas not overlain by ice, it exists beneath a layer of soil, rock or sediment.

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Desert Ecosystem

About 1/3rd of the worlds' land area is covered with desert

1. **Tropical Desert**
   - Notable features:
     - It is the driest and hottest place on earth.
     - Rainfall is sporadic and in some years no measurable precipitation falls at all.

   Examples: Sahara, Kalahari, Thar, Mexican deserts, Great Australian desert.
2. **Temperate Desert**
   - Notable features:
     - Temperate deserts can be much colder than tropical deserts
     - The floor of the temperate desert is often covered by rocks and small pebbles

   Examples: Mojave, Sonoran Deserts
3. **Cold Desert**
   - Notable features:
     - Cold deserts occur in temperate regions at higher latitudes
     - Hot summers but extremely cold winters.

   Examples: Atacama, Gobi, Great Basin, Namib, Iranian, Takla Makan, and Turkestan

Pond Ecosystem

Notable features:

- Small freshwater ecosystem
- Water is stagnant
- Can be seasonal
- Exposed to anthropogenic activities

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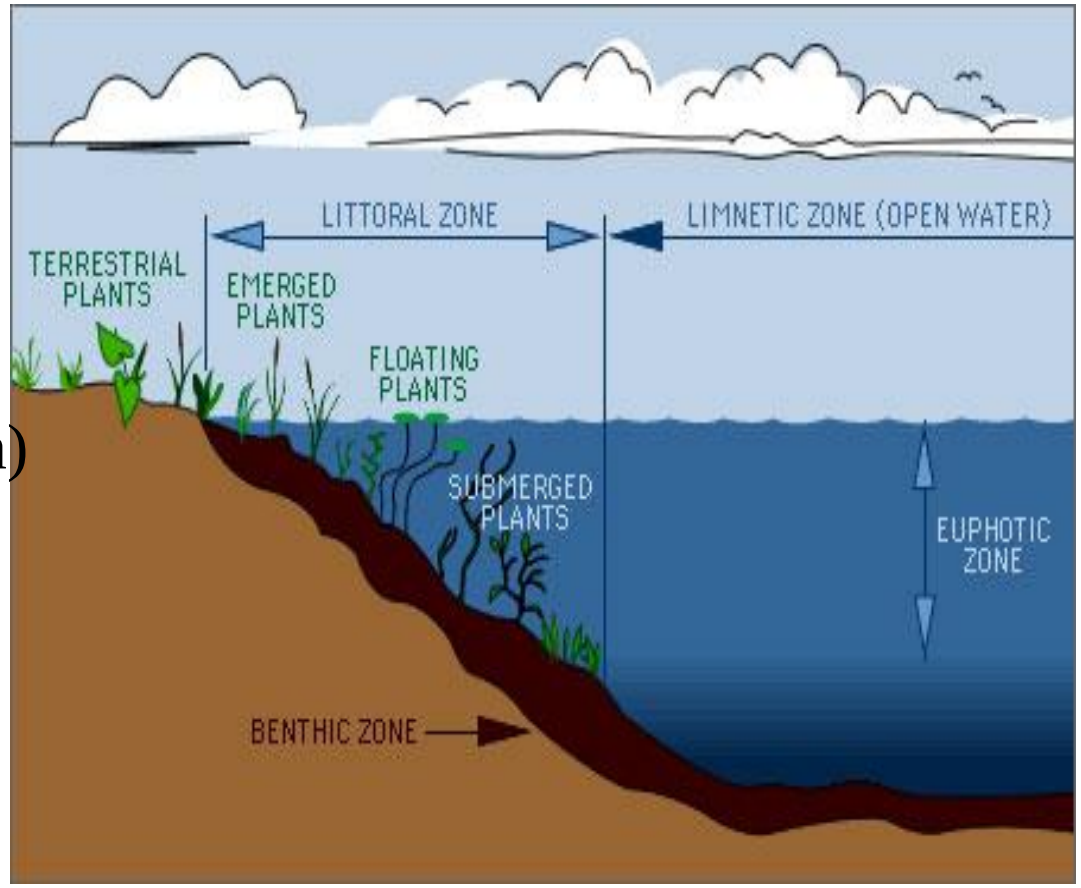
Lake Ecosystem



Organisms of aquatic ecosystem:

- Planktons (Algae, rotifers: limited locomotion)
- Nektons (Fishes: active swimmer)
- Neustons (Floating plants)
- Benthos (Bottom dweller: Snail)
- Periphytons (Crustaceane)

- Zonation (Stratification)
  - Epilimnion (Warm, lighter, circulating surface layer)
  - Hypolimnion (Cold, viscous, non-circulating bottom layer)

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Types of Lakes

- Oligotrophic lakes- Low nutrient content
- Eutrophic lake- High nutrient content
- Dystrophic lake- Low pH (Eg: **Lake** Suchar II in Poland, **lakes** Allgjuttern, Fiolen, and Brunnsjön in Sweden.)
- Endemic lakes- Ancient and deep having endemic fauna.
- Desert salt lakes- High salt content Eg: Great Salt Lake northern Utah, USA
- Volcanic lakes
- Mermictic lakes- Permanently stratified
- Artificial lakes

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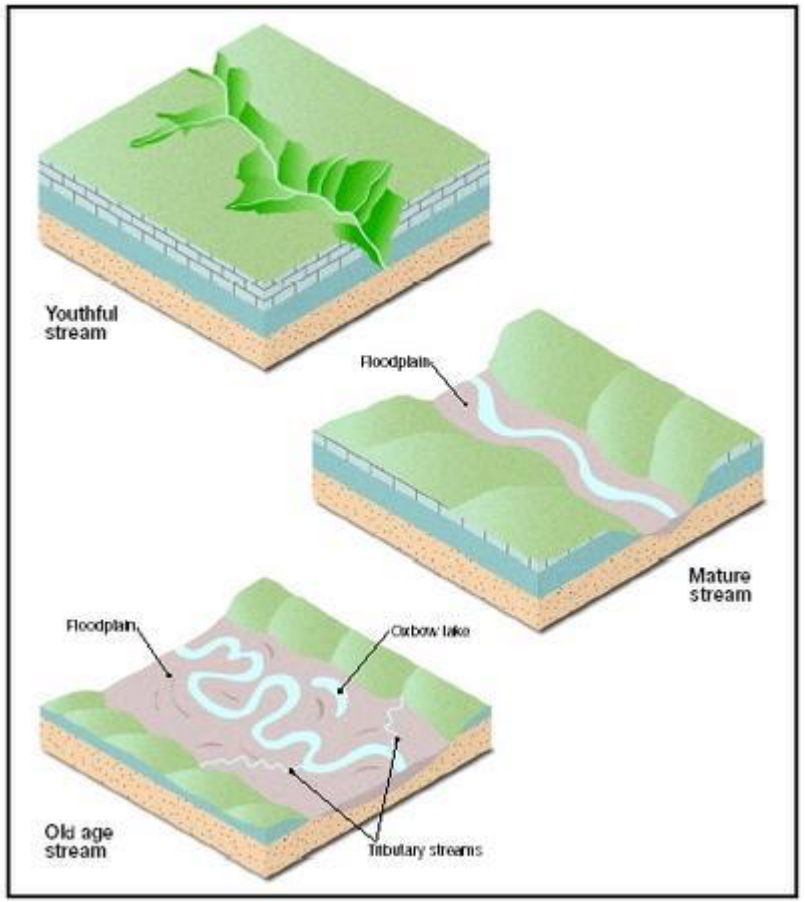
Streams

- Notable features:

  Stages

  - Mountain highland (Young River)
  - Second phase (Middle Aged River)
  - Third phase (Old River)



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Oceans Ecosystem

- Notable features:
  - Marine ecosystems support a great diversity of life and variety of habitats.
  - The ocean is a major influence on weather and climate.
  - Plants: seaweeds, or marine algae (brown, green, red), sea grasses, phytoplankton
  - Animals: protozoans, marine invertebrates (echinoderms, mollusks, segmented and non-segmented worms, jellies, coral, sea anemones, hyroids) marine vertebrates (fishes, birds, mammals), and zooplankton.
- Zones
  - Coastal zone
  - Open sea
    - Euphotic zone (Abundant sunlight, high photosynthetic activity)
    - Bathyal zone (Dim light)
    - Abyssal zone (Dark zone)

This material is for reference only. Points present in the slides are only for assistance and their elaboration are present in book. Examination questions will be in depth and can be solved by following text book only.